



Caringbah High School
 Year 12 2026
Mathematics Extension 2
 HSC Course
 Assessment Task 4 – Trial HSC Examination

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 100

Section I **10 marks**

Attempt Questions 1-10
 Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II **90 marks**

Attempt Questions 11-16
 Write your answers in the relevant section of the answer booklet provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only							
Section I	Section II						Total
Q 1-10	Q11	Q12	Q13	Q14	Q15	Q16	
/10	/15	/15	/15	/15	/15	/15	/100

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

1. What is the value of i^{767} ?

- (A) $-i$
- (B) -1
- (C) 1
- (D) i

2. Which of the following is the negation of the statement:

‘All solutions to the equation lie below the real axis on the complex plane.’

- (A) All solutions to the equation lie above the real axis
- (B) At least one solution to the equation lies above the real axis
- (C) All solutions to the equation lie on or above the real axis
- (D) At least one solution to the equation lies on or above the real axis

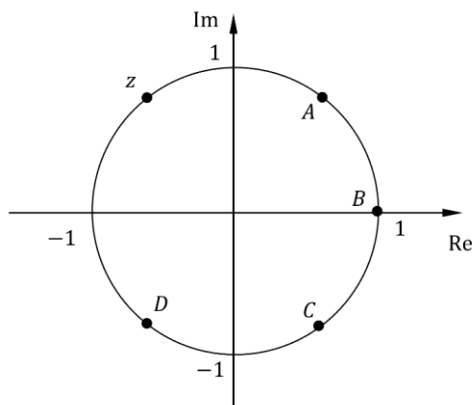
Section I continues next page

3. Consider the equation $z = 1 + i\sqrt{3}$.

Which of the following is equal to z^4 ?

- (A) $8 + 8i\sqrt{3}$
- (B) $8 - 8i\sqrt{3}$
- (C) $-8 + 8i\sqrt{3}$
- (D) $-8 - 8i\sqrt{3}$

4. A complex number z lies on the unit circle in the second quadrant of the complex plane, as shown in the diagram



Which of the points marked A , B , C and D best shows the position of $\bar{z}$?

- (A) A
- (B) B
- (C) C
- (D) D

Section I continues next page

5. The line $\vec{r} = \begin{pmatrix} 1 \\ -1 \\ \sqrt{7} \end{pmatrix} + \lambda \begin{pmatrix} \sqrt{7} \\ -2\sqrt{7} \\ c \end{pmatrix}$ is a tangent to a sphere centred at $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.

The point of contact between the tangent and the sphere is $\begin{pmatrix} 1 \\ -1 \\ \sqrt{7} \end{pmatrix}$.

What is the value of c ?

- (A) $c = -3$
 - (B) $c = -1$
 - (C) $c = 1$
 - (D) $c = 3$
6. Which of the following expressions would have an anti-derivative which includes $\sin^{-1}(f(x))$?

- (A) $\frac{2x}{\sqrt{x^4 - 2x^2 + 1}}$
- (B) $\frac{2x}{\sqrt{-x^4 - 4x^2 + 5}}$
- (C) $\frac{2x}{x^4 - 2x^2 + 1}$
- (D) $\frac{2x}{-x^4 - 4x^2 + 5}$

Section I continues next page

7. A particle is moving along the x -axis. At time t seconds it has displacement x metres from the origin O , velocity $v \text{ ms}^{-1}$, given by $v^2 = 10x - x^2$, and acceleration $a \text{ ms}^{-2}$. What is the value of $|a|$ when $|v| = 3$?
- (A) 1
(B) 2
(C) 3
(D) 4
8. Let $f(x)$ be a continuous function for all real x . It is given that $\int_0^\pi f(\sin x) dx = k$.
- What is the value of $\int_0^\pi x f(\sin x) dx$ in terms of k ?
- You may use the fact that $\int_0^a g(x) dx = \int_0^a g(a-x) dx$
- (A) 0
(B) $\frac{k}{2}$
(C) $\frac{\pi k}{2}$
(D) πk

Section I continues next page

9. A particle of unit mass is released from rest at $x = 1$ and moves in a straight line due to the force $F = \frac{5 - 7x}{x^3}$.

Which of the following equations represents the particle's velocity?

- (A) $V = \pm \frac{3}{x} \sqrt{x^2 - 7x + 5}$
- (B) $V = \pm \frac{1}{x} \sqrt{-9x^2 + 14x - 5}$
- (C) $V = \pm 3x \sqrt{x^2 - 7x + 5}$
- (D) $V = \pm x \sqrt{-9x^2 + 14x - 5}$

10. The complex number $z = x + iy$ satisfies $|z| \leq y + 1$ and $|z| \leq x + 1$.

What is the maximum possible value of $|z|$?

- (A) $3 + \sqrt{2}$
- (B) $1 + \sqrt{2}$
- (C) $2 - \sqrt{2}$
- (D) $2 + \sqrt{2}$

End of Section I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer the questions in the Section II Writing Booklet. Extra writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the **relevant section** in writing booklet.

(a) Write $1 - i\sqrt{3}$ in modulus-argument form. 2

(b) Consider the complex numbers $w = 4 - 3i$ and $z = 1 - 2i$.

(i) Evaluate $w + z$. 1

(ii) Evaluate $\frac{w}{z}$ in the form of $a + ib$. 2

(c) From an origin O , given vectors $\overrightarrow{OA} = \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$. 2

Find the angle between the two vectors, correct to the nearest minute.

(d) Find $\int xe^{5x} dx$. 2

(e) Prove by contradiction that $\sqrt{3} - \sqrt{2}$ is irrational. 2

(f) The vectors $\underline{u} = 2\underline{i} - 2\underline{j} + \underline{k}$ and $\underline{v} = a\underline{i} + b\underline{j} + c\underline{k}$ are parallel.

(i) Find $|\underline{u}|$. 1

(ii) Given that $|\underline{v}| = 6$, find the values of a , b and c . 3

End of Question 11

Question 12 (15 marks) Use the **relevant section** in writing booklet.

(a) Evaluate

2

$$\int \frac{2x-1}{x^2+2x+3} dx.$$

(b) A particle moves in simple harmonic motion along the x -axis. The central point of motion of the particle is the origin. The particle is initially at its right-hand extremity, 3 metres to the right of the origin, and takes 0.5 seconds to first reach the origin.

2

Find an equation for the displacement, x , of the particle in terms of t .

(c) Consider the function $f(x) = \frac{3kx-18}{(x+4)(x-2)}$, where k is a positive constant.

(i) Express $f(x)$ as partial fractions in terms of k .

2

(ii) Hence, find the exact value of k , such that $\int_{-3}^1 f(x)dx = 21$.

2

(d) A particle is moving in simple harmonic motion given by

$$\ddot{x} = -2\pi^2(x+1)$$

Initially the particle is at the centre of motion with velocity $3\pi\sqrt{2} \text{ ms}^{-1}$.

(i) Find the equation of v^2 .

2

(ii) Find the particle's exact period **and** the values of x at the endpoints of motion.

2

(e) For $n > 0$ fixed consider distinct constants $a, b, c \in (n, 2n)$.

(i) Show that $a + b - c > 0$.

1

(ii) Hence, or otherwise, show that $a > \sqrt{a^2 + 2cb - b^2 - c^2}$.

2

End of Question 12

Question 13 (15 marks) Use the **relevant section** in writing booklet.

- (a) For all $k \in \mathbb{Z}$, let P be the proposition: 2

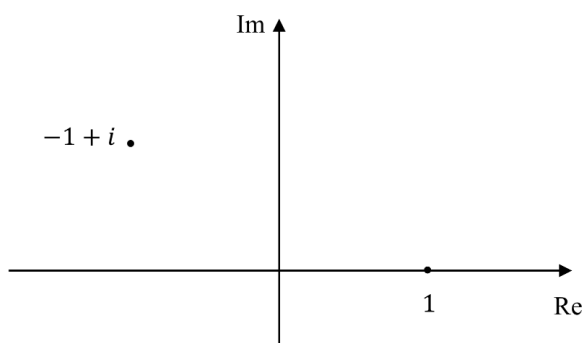
‘If $5k^2 + 3k + 4$ is odd, then k is even.’

Prove by contrapositive that P is true.

- (b) A particle starts at the origin at rest with acceleration, a , given by $a = -v^2 + 2v$, 3
where v is the velocity of the particle.

Find the expression for v in terms of x , the displacement of the particle.

- (c) A cubic polynomial, $P(z) = z^3 + bz^2 + cz + d$, where b, c and d are real numbers, has three distinct roots. Two of the roots are shown on the complex plane below.



- (i) What is the third root of $P(z)$? 1
(ii) Find the values of b, c and d . 3
(iii) What are the roots of $P(z - 1)$? 1

- (d) Show that $\int e^{6x} \sin(e^{3x}) dx = \frac{\sin(e^{3x}) - e^{3x} \cos(e^{3x})}{3} + c$ 2

Question 13 continues next page

(e) Let $f(x) = \ln\left(\frac{1+x}{1-x}\right) - 2x - \frac{2x^3}{3}$.

3

Use differentiation to prove that $\ln\left(\frac{1+x}{1-x}\right) > 2x + \frac{2x^3}{3}$ for $0 < x < 1$.

End of Question 13

Question 14 (15 marks) Use the **relevant section** in writing booklet.

- (a) Prove using mathematical induction that $a_n = \frac{n-1}{n+1}$ satisfies the recurrence relation 3

$$a_{n+1} = a_n + \frac{2}{n^2 + 3n + 2}$$

with the initial condition $a_1 = 0$, for all integers $n \geq 1$.

- (b) Using a suitable substitution, show that

$$\int_0^{\ln(2)} \frac{e^{2x}}{\sqrt{e^x + 1}} dx = \frac{2\sqrt{2}}{3} \quad 3$$

- (c) Sketch the region of the complex plane consisting of all $z \in \mathbb{C}$ such that 3

$$|\text{Arg}(z^2)| < \frac{\pi}{6}$$

- (d) (i) Use De Moivre's theorem to show that 2

$$\cos(4\theta) = 8\cos^4(\theta) - 8\cos^2(\theta) + 1.$$

- (ii) Hence, or otherwise, show that the equation $16x^4 - 16x^2 + 1 = 0$ has 2

the roots $x = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos \frac{7\pi}{12}$ and $\cos \frac{11\pi}{12}$.

- (iii) Hence, show that $\cos \frac{5\pi}{12} = \frac{\sqrt{2-\sqrt{3}}}{2}$. 2

End of Question 14

Question 15 (15 marks) Use the **relevant section** in writing booklet.

- (a) The point of intersection of the line through $A \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ and $B \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix}$ with the line

through $C \begin{pmatrix} 0 \\ 9 \\ 10 \end{pmatrix}$ and $D \begin{pmatrix} 1 \\ 7 \\ 7 \end{pmatrix}$ is E .

- (i) Prove that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$. 1
- (ii) Find the coordinates of E . 2

- (b) Find $\int x \cos^{-1} x \, dx$. 3

- (c) A particle is projected from the origin with initial velocity u at an angle α to the horizontal. The particle experiences a resistive force proportional to its velocity in both the horizontal and vertical directions, such that the equations of motion are: 3

$$\ddot{x} = -k\dot{x} \text{ and } \ddot{y} = -k\dot{y} - g$$

where k is a positive constant and g is the acceleration due to gravity.

Let θ be the angle that the trajectory makes with the horizontal at any time t .

Prove that

$$\tan \theta = \tan \alpha - \frac{g(e^{kt} - 1)}{ku \cos \alpha}.$$

- (d) Let $I_n = \int \sec^n(x) dx$ for $n \geq 0$.

- (i) For $n \geq 2$, show that $I_n = \frac{\tan(x) \sec^{n-2}(x)}{n-1} + \frac{n-2}{n-1} I_{n-2}$. 3
- (ii) Hence, using an appropriate substitution, show that 3

$$\int_1^2 \sqrt{x^2 - 1} \, dx = \sqrt{3} - \frac{1}{2} \ln(2 + \sqrt{3})$$

End of Question 15

Question 16 (15 marks) Use the **relevant section** in writing booklet.

(a) Prove by mathematical induction that

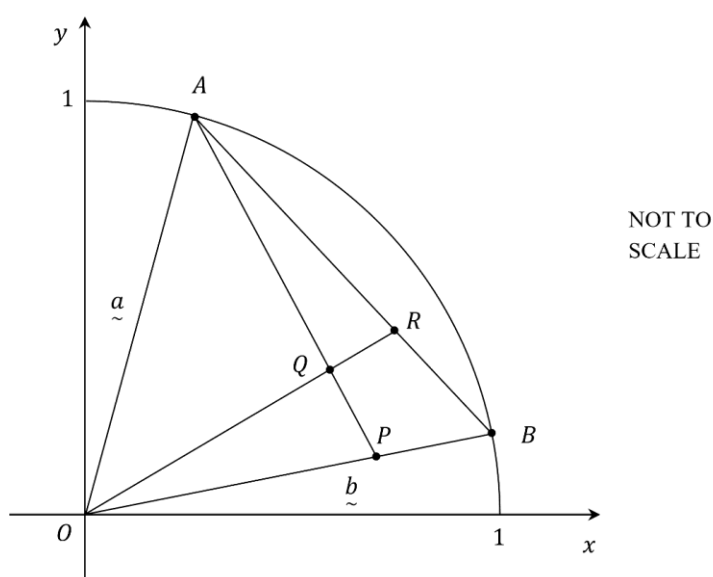
3

$$\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{n} \geq \frac{2}{3}n\sqrt{n}, \text{ for all } n \geq 1$$

(b) In triangle OAB , A and B lie on the unit circle in the first quadrant as shown.

P lies on OB such that $\overrightarrow{OP} = 3\overrightarrow{PB}$, Q lies on AP such that $\overrightarrow{AQ} = 2\overrightarrow{QP}$, and

R lies on AB such that O, Q and R are collinear. Let $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$.



(i) It is given $\overrightarrow{AR} = k\overrightarrow{AB}$. Find the value of k .

3

(ii) Given that $\overrightarrow{OQ} \cdot \overrightarrow{AP} = 0$, find the size of $\angle AOB$, correct to the nearest minute.

2

(c) A particle of unit mass is projected vertically upwards from the ground with an initial speed of u .

3

It experiences a resistive force, kv , and a gravitational force, g .

When the particle lands back on the ground, it has a speed of w .

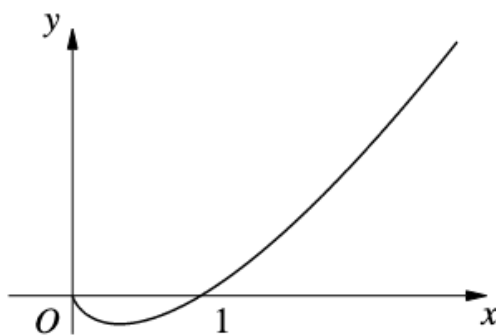
Show that the particle takes **in total** $T = \frac{u + w}{g}$ seconds to land back on the

ground.

Question 16 continues next page

- (d) Define the sequence a_n by $a_1 = 1$ and $a_{k+1} = a_k \times (k+1)^{k+1}$.

4



By considering the area under the curve $y = x \ln x$ shown above, prove that

$$a_n > e^{\frac{n^2}{4}(2\ln(n)-1)}.$$

End of Examination

Solutions

MC

$$\begin{aligned} 1) \quad i^{167} &= (i^4)^{41} \times i^3 \\ &= i^{164} \times (-i) \\ &= -i \end{aligned} \quad \therefore (A)$$

2) (D)

$$\begin{aligned} 3) \quad (1+i\sqrt{3})^4 &= \left[2 \left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right) \right) \right]^4 \\ &= 16 \left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3} \right) \quad \therefore (D) \\ &= 16 \left(-\frac{1}{2} + \left(-\frac{i\sqrt{3}}{2}\right) \right) \\ &= -8 - 8i\sqrt{3} \end{aligned}$$

$$4) \quad z\bar{z} = |z|^2 = 1 \quad \therefore (B)$$

5) $r \perp \text{tangent}$:

$$\left(\frac{1}{\sqrt{7}} \right) \cdot \left(\frac{\sqrt{7}}{-2\sqrt{7}} \right) = 0 \quad \therefore (A)$$

$$\sqrt{7} + 2\sqrt{7} + c\sqrt{7} = 0$$

$$3\sqrt{7} + c\sqrt{7} = 0$$

$$c = -3$$

6) The integrand has to be written in form

$$\frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}}$$

(B)

A: $\frac{2x}{\sqrt{(x^2-1)^2}}$ X

B: $\frac{2x}{\sqrt{-(x^4+4x^2+4-4)+5}} = \frac{2x}{\sqrt{-(x^2+2)^2+9}}$
 $= \frac{2x}{\sqrt{9-(x^2+2)^2}}$

C: X

D: X

7) $a = \frac{1}{2} \frac{d}{dx}(x^2)$
 $a = \frac{1}{2}(10-2x)$
 $a = 5-x$

$|v| = 3 \Rightarrow |v|^2 = 9$
 $-x^2 + 10x = 9$
 $x^2 - 10x + 9 = 0$
 $(x-9)(x-1) = 0$
 $x = 9$ or $x = 1$

When $x = 9 \Rightarrow a = 4$
 $x = 1 \Rightarrow a = -4$ } $\Rightarrow |a| = 4$ (D)

8) Let $I = \int_0^n x f(\sin x) dx$

Since $\int_0^a g(x) dx = \int_0^a g(a-x) dx$

$$I = \int_0^n (n-x) f(\sin(n-x)) dx$$

* $\sin(n-x) = \sin x$

s. $I = \int_0^n (n-x) f(\sin x) dx$

$$I = n \int_0^n f(\sin x) dx - \int_0^n x f(\sin x) dx$$

$$I = nk - I$$

$$2I = nk$$

$$I = \frac{nk}{2}$$

(C)

9) $f = mc$, unit mass $\Rightarrow m = 1$
so $a = f$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{5-7x}{x^3}$$

$$\frac{1}{2} v^2 = \frac{5x^2}{-2} - \frac{7x^1}{-1}$$

$$v^2 = -\frac{5}{x^2} + \frac{14}{x} + C$$

at $v=0$, $x=1$

$$0 = -5 + 14 + C$$

$$\Rightarrow C = -9$$

$$\Rightarrow V^2 = -\frac{5}{x^2} + \frac{14}{x} - 9$$

$$V^2 = -\frac{5 + 14x - 9x^2}{x^2}$$

∴ (B)

$$V = \pm \frac{1}{x} \sqrt{-9x^2 + 14x - 5}$$

10) Multiple ways to do this;

the inequality is symmetric
in x & y , so max occurs
when $x=y$

$$\begin{aligned} |z| &= \sqrt{x^2 + x^2} \\ &= \sqrt{2x^2} \end{aligned}$$

$$\sqrt{2x^2} = x+1$$

$$2x^2 = x^2 + 2x + 1$$

$$x^2 - 2x - 1 = 0$$

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2}$$

$$= \frac{2 \pm \sqrt{2}}{2}$$

$$x = 1 \pm \sqrt{2}$$

$$\Rightarrow y = 1 \pm \sqrt{2}$$

$$|z|_{\max} \text{ when } x=y=1+\sqrt{2} \text{ as } 1+\sqrt{2} > 1-\sqrt{2}$$

$$\begin{aligned} |z| &= \sqrt{2}(1+\sqrt{2}) \\ &= 2+\sqrt{2} \end{aligned}$$

$$|z| \leq y+1$$

$$|z| \leq x+1$$

$$y \geq |z|-1$$

$$x \geq |z|-1$$

$$x^2 + y^2 \geq 2(|z|-1)^2$$

$$|z|^2 \geq 2 \left(\frac{|z|^2 - 2|z| + 2}{2(|z|+1)} \right)$$

$$|z|^2 - 4|z| + 2 \leq 0$$

$$|z| = \frac{4 \pm \sqrt{16 - 4(2)}}{2}$$

$$(D) \left\{ \begin{aligned} |z| &= 2 \pm \sqrt{2} \end{aligned} \right.$$

Max $\Rightarrow$

$$|z| = 2 + \sqrt{2}$$

(D)

Question 11

$$a) \quad \text{Li3} = 2 \left(\underset{\textcircled{1}}{\cos\left(-\frac{\pi}{3}\right)} + \underset{\textcircled{1}}{\sin\left(-\frac{\pi}{3}\right)} \right)$$

$$b) \quad i) \quad w+z = 5-5i$$

$$ii) \quad \frac{w}{z} = \frac{(4-3i)(4+2i)}{(1-2i)(1+2i)} - \textcircled{1}$$

$$= \frac{4+8i-3i+6}{1+4}$$

$$= \frac{10+5i}{5}$$

$$= 2+i - \textcircled{1}$$

$$c) \quad \cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|}$$

$$= \frac{-1-6+3}{\sqrt{11} \times \sqrt{14}}$$

$$= \frac{-4}{\sqrt{154}} - \textcircled{1}$$

$$\theta \approx 108^\circ 48' - \textcircled{1}$$

$$d) \quad \int x e^{5x} dx$$

$$u = x$$

$$u' = 1$$

$$v' = e^{5x}$$

$$v = \frac{e^{5x}}{5}$$

$$= \left[\frac{x e^{5x}}{5} \right] - \frac{1}{5} \int e^{5x} dx - \textcircled{1}$$

$$= \frac{1}{5} x e^{5x} - \frac{1}{25} e^{5x} + C \quad \text{--- (1)}$$

e) Assume $\sqrt{3} - \sqrt{2}$ is rational

so $\exists m, n \in \mathbb{N}$ coprime such that

$$\sqrt{3} - \sqrt{2} = \frac{m}{n}$$

$$\sqrt{3} = \frac{m}{n} + \sqrt{2}$$

$$3 = \frac{m^2}{n^2} + \frac{2\sqrt{2}m}{n} + 2$$

$$1 - \frac{m^2}{n^2} = \frac{2\sqrt{2}m}{n} \quad \text{--- (1)}$$

$$2\sqrt{2} = \frac{n}{m} - \frac{m^2 n}{n^2 m}$$

$$\sqrt{2} = \frac{n^2 - m^2}{2mn} \in \mathbb{Z}, \quad \in \mathbb{Q}$$

but $\sqrt{2}$ is irrational so contradiction --- (1)

$$p) i) |u| = \sqrt{4+4+1} = 3 \quad \text{--- (1)}$$

$$ii) |v| = 6 = 2|u| \quad \text{--- (1)}$$

$$\Rightarrow v = \pm 2u$$

$$= \pm 2(2i - 2j + k) \quad \text{--- (1)}$$

$$= \pm (4i - 4j + 2k) \quad \dots$$

$$v = 4i - 4j + 2k \quad \text{or} \quad v = -4i + 4j - 2k \quad \text{--- (1)}$$

Question 12

$$a) \int \frac{2x-1}{x^2+2x+3} dx$$

$$= \int \frac{2x+2-3}{x^2+2x+3} dx$$

$$= \int \frac{2x+2}{x^2+2x+3} dx - \int \frac{3}{x^2+2x+3} dx$$

$$= \ln|x^2+2x+3| - 3 \int \frac{1}{(x+1)^2+2} dx$$

$$= \ln|x^2+2x+3| - \frac{3}{\sqrt{2}} \tan^{-1} \left| \frac{x+1}{\sqrt{2}} \right| + C$$

①

①

$$b) T = 4 \times 0.5$$

$$= 2s$$

$$\frac{2n}{n} = 2 \Rightarrow n = \pi$$

①

* starts extreme right $\Rightarrow x = A \cos(\omega t)$

$$A = 3 - 0 = 3$$

$$\boxed{x = 3 \cos \pi t} \quad \text{--- ①}$$

$$c) i) \int \frac{3kx-18}{(x+4)(x-2)} = \frac{a}{x+4} + \frac{b}{x-2}$$

$$\frac{3kx-18}{(x+4)(x-2)} = \frac{a(x-2) + b(x+4)}{(x+4)(x-2)}$$

$$3kx-18 = a(x-2) + b(x+4)$$

$$x \rightarrow 2 \quad 6k-18 = 6b$$

$$b = k-3 \quad \text{--- (1)}$$

$$x \rightarrow -4 \quad 12k-18 = -6a$$

$$a = 2k+3 \quad \text{--- (2)}$$

$$f(x) = \frac{2k+3}{x+4} + \frac{k-3}{x-2}$$

$$ii) \int_{-3}^1 f(x) dx = 21$$

$$\int_{-3}^1 \left(\frac{2k+3}{x+4} + \frac{k-3}{x-2} \right) dx = 21$$

$$\left[(2k+3) \ln|x+4| + (k-3) \ln|x-2| \right]_{-3}^1 = 21 \quad \text{--- (1)}$$

$$(2k+3) \ln 5 + (k-3) \ln 1 - (2k+3) \ln 1 - (k-3) \ln 5 = 21$$

$$(k+6) \ln 5 = 21$$

$$k = \frac{21}{\ln 5} - 6 \quad \text{--- (2)}$$

-8-

$$d) i) \frac{d}{dn} \left(\frac{1}{2} v^2 \right) = -2n^2 (n+1)$$

$$\frac{1}{2} v^2 = -2n^2 \left(\frac{n^2}{2} + n \right) + C$$

$$v^2 = -4n^2 \left(\frac{n^2}{2} + n \right) + C_1$$

$$v^2 = -2n^4 n^2 - 4n^3 n + C_1 \quad \text{--- (1)}$$

$$n = -1, v = 3n\sqrt{2}$$

$$18n^2 = -2n^2(1) - 4n^3(-1) + C_1$$

$$C_1 = 18n^2 - 2n^2$$

$$C_1 = 16n^2$$

$$v^2 = -2n^4 n^2 - 4n^3 n + 16n^2 \quad \text{--- (1)}$$

$$ii) v^2 = -2n^2 (n^2 + 2n + 8)$$

$$v^2 = -2n^2 (n+4)(n-2)$$

$$n^2 = 2n^2$$

$$n = \sqrt{2}n$$

$$P = \frac{2n}{\sqrt{2}n} = \sqrt{2} \quad \text{--- (1)}$$

$$n+4=0 \text{ or } n-2=0$$

$$\boxed{n=-4} \text{ or } \boxed{n=2}$$

$$\text{--- (1)}$$

e) i) $a > n$
 $b > n$
 $c < 2n \Rightarrow -c > -2n$

$$a + b - c > n + n - 2n \quad \text{--- (1)}$$

$$a + b - c > 0$$

ii) by same logic of i)
 $a + c - b > 0$

by AM-GM

$$\frac{(a+b-c)(a+c-b)}{2} > \sqrt{(a+b-c)(a+c-b)} \quad \text{--- (1)}$$

$$a > c - b \text{ \& } a > b - c$$

so

$$\frac{2a}{2} > \sqrt{a^2 - (b-c)^2}$$

$$a > \sqrt{a^2 - b^2 - c^2 + 2bc} \quad \text{--- (1)}$$

Question 13

a) If k is odd then $5k^2+3k+4$ is even — (1)

$$\text{Let } k=2m+1, m \in \mathbb{Z}$$

$$\begin{aligned} 5k^2+3k+4 &= (2m+1)^2 + (2m+1) + 4 \\ &= 5(4m^2+4m+1) + 6m+7 \\ &= 20m^2 + 26m + 12 \\ &= 2(10m^2 + 13m + 6) \quad \text{--- (1)} \\ &\quad \underbrace{\hspace{1.5cm}}_{\in \mathbb{Z}} \end{aligned}$$

$\Rightarrow$ even

b) $a = -v^2 + 2v$

$$v \frac{dv}{dn} = -v^2 + 2v$$

$$\frac{dv}{dn} = -v + 2$$

$$\frac{dn}{dv} = \frac{1}{-v+2} \quad \text{--- (1)}$$

$$\int_0^n \frac{dn}{dv} = \int_0^v \frac{dv}{-v+2}$$

$$n = -(\ln|-v+2|) \Big|_0^v$$

$$n = (\ln(2) - \ln(-v+2))$$

$$\ln(-v+2) = \ln 2 - n \quad \text{--- (1)}$$

$$-v + 2 = 2e^{-u}$$

$$v = 2 - 2e^{-u}$$

$$v = 2(1 - e^{-u}) \quad \text{--- (1)}$$

c) i) $\overline{-1+i} = -1-i$

ii) $1 - 1 + i - 1 - i = -b$

$$\boxed{b=1} \quad \text{--- (1)}$$

$$\alpha\beta + \beta\gamma + \beta\alpha = \frac{c}{a}$$

$$(-1+i) + (-1-i) + (-1-i)(-1+i) = c$$

$$-2 + (-1)^2 - (i)^2 = c$$

$$-1 + 1 = c \Rightarrow \boxed{c=0} \quad \text{--- (1)}$$

$$(1)(-1+i)(-1-i) = -\frac{d}{a}$$

$$\boxed{d=-2} \quad \text{--- (1)}$$

iii) $P(z-1) = \pm i, 2 \quad \left(\begin{smallmatrix} \text{one} \\ \text{unit} \\ \text{right} \end{smallmatrix} \right) \quad \text{--- (1)}$

d) next page.

$$d) \int e^{6x} \sin(e^{3x}) = \frac{1}{3} \int 3e^{3x} \times e^{3x} \sin(e^{3x}) dx$$

$$u = e^{3x}$$

$$v' = 3e^{3x} \sin(e^{3x})$$

① -

$$u' = 3e^{3x}$$

$$v = -\cos(e^{3x})$$

$$= \frac{1}{3} \left([-e^{3x} \cos(e^{3x})] + \int 3e^{3x} \cos(e^{3x}) dx \right)$$

$$= -\frac{1}{3} e^{3x} (\cos(e^{3x})) + \frac{1}{3} \sin e^{3x} + C$$

$$= \frac{\sin(e^{3x}) - e^{3x} \cos(e^{3x})}{3} + C \quad \text{--- ①}$$

$$e) f'(x) = \frac{1}{1+x} + \frac{1}{1-x} \quad -2 - 2x^2$$

$$= \frac{1-x + 1+x - 2(1-x^2) - 2x - (1-x)}{(1+x)(1-x)}$$

$$= \frac{2 - 2 + 2x^2 - 2x^2 + 2x^4}{1-x^2}$$

$$= \frac{2x^4}{1-x^2} \quad \text{--- ①}$$

$$f''(x) = 0$$

$$\Rightarrow x = 0$$

x	$-\frac{1}{6}$	0	$\frac{1}{6}$
y'	$\frac{1}{6}$	$-$	$\frac{1}{6}$
	$\swarrow$	$-$	$\swarrow$

$$\Rightarrow f(x) > 0 \quad 0 < x < 1$$

$$\left(n \left(\frac{1+x}{1-x} \right) > 2x + \frac{2x^3}{3} \right) \quad 0 < x < 1 \quad \text{--- ①}$$

Question 14

$$a) \quad n=1 \quad a_1 = \frac{1-1}{2} = 0 \quad \checkmark$$

Assume true for $n=k$, so $a_k = \frac{k-1}{k+1}$

To prove true for $n=k+1$

$$\text{that is } a_{k+1} = \frac{k+1-1}{k+1+1} = \frac{k}{k+2}$$

$$\text{LHS } a_{k+1} = \frac{k-1}{k+1} + \frac{2}{k^2+3k+2}$$

$$= \frac{k-1}{k+1} + \frac{2}{(k+1)(k+2)}$$

$$= \frac{(k-1)(k+2) + 2}{(k+1)(k+2)}$$

$$= \frac{k^2 + k - 2 + 2}{(k+1)(k+2)}$$

$$= \frac{k(k+1)}{(k+1)(k+2)}$$

$$= \frac{k}{k+2} \quad \checkmark$$

$$b) \int_0^{\ln 2} \frac{e^{2x}}{\sqrt{e^x + 1}} dx = \frac{2\sqrt{2}}{3}$$

$$\begin{aligned} \text{let } u &= e^x + 1 \\ \frac{du}{dx} &= e^x \Rightarrow du = e^x dx \end{aligned} \quad (1)$$

$$= \int_2^3 \frac{(u-1)}{\sqrt{u}} du$$

$$= \int_2^3 \left(\frac{u}{\sqrt{u}} - \frac{1}{\sqrt{u}} \right) du$$

$$= \left[\frac{2}{3} u^{\frac{3}{2}+1} - 2u^{-\frac{1}{2}+1} \right]_2^3 \quad (1)$$

$$= \left(\frac{2 \times 3\sqrt{3}}{3} - 2\sqrt{3} \right) - \left(\frac{2 \times 2\sqrt{2}}{3} - 2\sqrt{2} \right)$$

$$= (2\sqrt{3} - 2\sqrt{3}) - \left(\frac{4\sqrt{2}}{3} - 2\sqrt{2} \right)$$

$$= -\frac{4\sqrt{2} + 6\sqrt{2}}{3} \quad (1)$$

$$= \frac{2\sqrt{2}}{3}$$

$$c) \quad |Arg(z^2)| < \frac{\pi}{6}$$

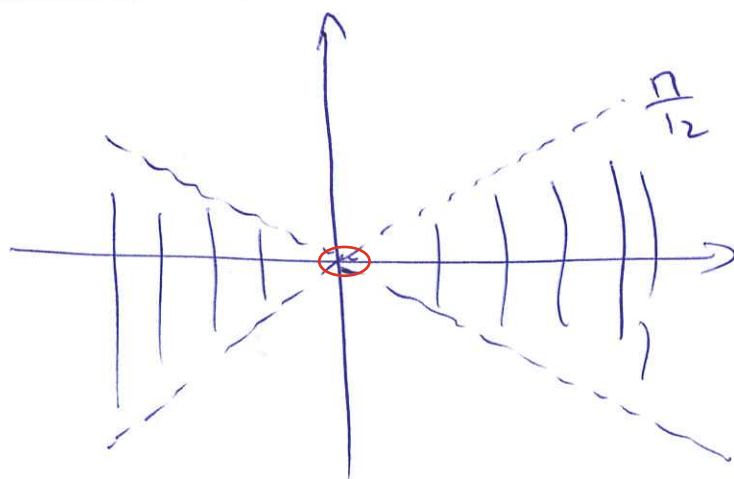
$$-\frac{\pi}{6} < Arg(z^2) < \frac{\pi}{6}$$

$$\text{so } -\frac{\pi}{6} < 2\theta < \frac{\pi}{6} \quad \text{--- (1)}$$

$$\Rightarrow -\frac{\pi}{12} < \theta < \frac{\pi}{12}$$

& to cover $2\theta + 2kn$ in $(-\frac{\pi}{6}, \frac{\pi}{6})$

$$-\frac{\pi}{12} \leq \theta \leq \frac{\pi}{12} \quad \& \quad \frac{21\pi}{12} \leq \theta \leq \pi \quad \& \quad -\pi < \theta < -\frac{11\pi}{12} \quad \text{--- (1)}$$



--- (1)

$$d) i) (\cos\theta + i\sin\theta)^4 = \cos^4\theta + i\sin^4\theta$$

$$(\cos\theta + i\sin\theta)^4 = \cos^4\theta - 4\cos^3\theta\sin\theta i - 6\cos^2\theta\sin^2\theta - 4\cos\theta\sin^3\theta i + \sin^4\theta \quad \text{--- (1)}$$

$$\begin{aligned} \Rightarrow \cos^4\theta &= \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta \\ &= \cos^4\theta - 6\cos^2\theta(1-\cos^2\theta) + (1-\cos^2\theta)^2 \\ &= \cos^4\theta - 6\cos^2\theta + 6\cos^4\theta + 1 - 2\cos^2\theta + \cos^4\theta \end{aligned} \quad \text{--- (1)}$$

$$= 8\cos^4\theta - 8\cos^2\theta + 1$$

$$\text{ii) } 16x^4 - 16x^2 + 1 = 0$$

$$2(8x^4 - 8x^2 + 1) - 2 + 1 = 0$$

$$2(8x^4 - 8x^2 + 1) = 1$$

$$8x^4 - 8x^2 + 1 = \frac{1}{2} \quad \text{--- (1)}$$

$$\text{Let } x = \cos\theta$$

$$8\cos^4\theta - 8\cos^2\theta + 1 = \frac{1}{2}$$

$$\cos^4\theta = \frac{1}{2}$$

$$\cos\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12} \quad \text{--- (1)}$$

$$\text{iii) } x^2 = \frac{16 \pm \sqrt{16^2 - 4 \times 16}}{32}$$

$$x^2 = \frac{16 \pm \sqrt{192}}{32}$$

$$= \frac{2 \pm \sqrt{3}}{4} \quad \text{--- (1)}$$

$$\frac{5\pi}{12} \text{ is in Q1 so } \cos\left(\frac{5\pi}{12}\right) > 0$$

$$\Rightarrow (\cos\theta)^2 = \frac{2 \pm \sqrt{3}}{4}$$

$$\cos\theta = \sqrt{\frac{2 \pm \sqrt{3}}{4}}$$

$$\cos\theta = \frac{\sqrt{2 \pm \sqrt{3}}}{2}$$

Question 15

$$a) i) \vec{AB} = \begin{bmatrix} 3 \\ 6 \\ -3 \end{bmatrix} \quad \vec{CD} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

$$\vec{AB} \cdot \vec{CD} = 3 - 12 + 9 \\ = 0$$

b) ii) $r_{AB} = \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} 3 \\ 6 \\ -3 \end{bmatrix}$ → can use A instead

$r_{CD} = \begin{bmatrix} 0 \\ 9 \\ 10 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ → can use D instead

$$\begin{cases} 4 + 3\lambda = \mu & \text{--- (1)} \\ 5 + 6\lambda = 9 - 2\mu & \text{--- (2)} \\ 0 - 3\lambda = 10 - 3\mu & \text{--- (3)} \end{cases}$$

————— (1)

① - ③ $4 = -2\mu + 10$

$$-2\mu = -6$$

$$\mu = 3$$

sub in ①

$$3\lambda = 3 - 4$$

$$\lambda = -\frac{1}{3}$$

check in ② $5 + 6\left(-\frac{1}{3}\right) = 5 - 2 = 3 \quad \checkmark$

$$9 - 2\mu = 9 - 6 = 3 \quad \checkmark$$

$$\Rightarrow E = \begin{bmatrix} 0 + 3 \\ 9 - 6 \\ 10 - 9 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}$$

————— (1)

$$b) \int x \cos^{-1} x \, dx$$

$$u = \cos^{-1} x$$

$$v' = x$$

$$u' = \frac{-1}{\sqrt{1-x^2}}$$

$$v = \frac{x^2}{2}$$

$$= \left[\frac{x^2}{2} \cos^{-1} x \right] + \int \frac{x^2}{2\sqrt{1-x^2}} \, dx \quad \text{--- (1)}$$

$$* \int \frac{x^2}{2\sqrt{1-x^2}} \, dx \quad \text{Let } x = \sin \theta$$

$$dx = \cos \theta \, d\theta$$

$$= \frac{1}{2} \int \frac{\sin^2 \theta \cos \theta \, d\theta}{1 - \sin^2 \theta}$$

$$= \frac{1}{2} \int \sin^2 \theta \, d\theta$$

$$= \frac{1}{4} \int (1 - \cos 2\theta) \, d\theta$$

$$= \frac{1}{4} \theta - \frac{1}{8} \sin 2\theta + C \quad \text{--- (1)}$$

$$* \int x \cos^{-1} x \, dx = \left[\frac{x^2}{2} \cos^{-1} x \right] + \frac{1}{4} \theta - \frac{1}{8} \sin 2\theta + C$$

$$= \frac{x^2}{2} \cos^{-1} x + \frac{1}{4} \sin^{-1} x - \frac{1}{4} \sin(\sin^{-1} x) \cos(\sin^{-1} x) + C$$

$$= \frac{x^2}{2} \cos^{-1} x + \frac{1}{4} \sin^{-1} x - \frac{1}{4} x \sqrt{1-x^2} + C \quad \text{--- (1)}$$

$$c) \tan \theta = \frac{\dot{y}}{\dot{x}}$$

$$\ddot{x} = -kx$$

exp decay so

$$\dot{x} = Ae^{-kt}$$

at $t=0$, $\dot{x} = u \cos \alpha$

$$\dot{x} = u \cos \alpha e^{-kt}$$

$$\ddot{y} = -ky - g$$

exp decay

$$y = Ae^{-kt} - \frac{g}{k}$$

at $t=0$, $y = u \sin \alpha$

$$\dot{y} = \left(u \sin \alpha + \frac{g}{k} \right) e^{-kt} - \frac{g}{k}$$

①

$$\tan \theta = \frac{\dot{y}}{\dot{x}}$$

$$= \frac{u \sin \alpha e^{-kt} + \frac{g}{k} e^{-kt} - \frac{g}{k}}{u \cos \alpha e^{-kt}}$$

$$\tan \theta = \frac{k u \sin \alpha e^{-kt} + g e^{-kt} - g}{k u \cos \alpha e^{-kt}} \quad \text{--- ①}$$

$$\tan \theta = \frac{\cancel{k u \sin \alpha e^{-kt}} + \frac{g e^{-kt} - g}{k u \cos \alpha e^{-kt}}}{\cancel{k u \cos \alpha e^{-kt}}}$$

$$\tan \theta = \tan \alpha + \frac{g(e^{-kt} - 1)}{k u \cos \alpha e^{-kt}} \times e^{kt}$$

$$\tan \theta = \tan \alpha + \frac{g(1 - e^{kt})}{k u \cos \alpha}$$

$$\tan \theta = \tan \alpha - \frac{g(e^{kt} - 1)}{k u \cos \alpha} \quad \text{--- ①}$$

$$d) I_n = \int \sec^n(x) dx$$

$$= \int \sec^{n-2}(x) \sec^2(x) dx$$

$$u = \sec^{n-2}(x)$$

$$v' = \sec^2(x) \quad \text{--- (1)}$$

$$u' = (n-2) \sec^{n-3}(x) [\sec x \tan x] \quad v = \tan x$$

$$I_n = \tan x \sec^{n-2}(x) - (n-2) \int \sec^{n-2}(x) \tan^2 x dx$$

$$I_n = \tan x \sec^{n-2}(x) - (n-2) \int \sec^{n-2}(x) [\sec^2 x - 1] dx \quad \text{--- (1)}$$

$$I_n = \tan x \sec^{n-2}(x) - (n-2) \left[\int \sec^2 x dx - \int \sec^{n-2}(x) dx \right]$$

$$I_n = \tan x \sec^{n-2}(x) - (n-2) [I_n - I_{n-2}]$$

$$\underline{I_n + (n-2) I_n} = \tan x \sec^{n-2}(x) + (n-2) I_{n-2}$$

$$(n-1) I_n = \tan x \sec^{n-2}(x) + (n-2) I_{n-2}$$

$$I_n = \frac{\tan x \sec^{n-2}(x)}{n-1} + \frac{(n-2)}{n-1} I_{n-2} \quad \text{--- (1)}$$

ii) Let $x = \sec u$ * $x=1 \Rightarrow u=0$
 $dx = \sec u \tan u du$ * $x=2 \Rightarrow u = \frac{\pi}{3}$

$$2 \int \sqrt{x^2 - 1} dx = \int_0^{\frac{\pi}{3}} \sqrt{\sec^2 u - 1} (\sec u \tan u) du$$

$$= \int_0^{\frac{\pi}{3}} \sec u \sqrt{\tan^2 u} \tan u \, du$$

$$= \int_0^{\frac{\pi}{3}} \sec u \tan^2 u \, du$$

$$= \int_0^{\frac{\pi}{3}} \sec u [\sec^2 u - 1] \, du$$

$$= \int_0^{\frac{\pi}{3}} (\sec^3 u - \sec u) \, du$$

$$= \int_0^{\frac{\pi}{3}} \sec^3 u \, du - \int_0^{\frac{\pi}{3}} \sec u \, du \quad \text{--- (1)}$$

$$= \left[\frac{\tan u \sec u}{2} \right]_0^{\frac{\pi}{3}} + \frac{1}{2} \int_0^{\frac{\pi}{3}} \sec u \, du - \int_0^{\frac{\pi}{3}} \sec u \, du$$

$$= \left[\frac{\tan u \sec u}{2} \right]_0^{\frac{\pi}{3}} - \frac{1}{2} \int_0^{\frac{\pi}{3}} \sec u \, du \quad \text{--- (1)}$$

$$\therefore = \left[\frac{\tan u \sec u}{2} \right]_0^{\frac{\pi}{3}} - \frac{1}{2} \int_0^{\frac{\pi}{3}} \frac{\sec u (\tan u + \sec u)}{\tan u + \sec u} \, du$$

$$= \left[\frac{\tan u \sec u}{2} \right]_0^{\frac{\pi}{3}} - \frac{1}{2} \left[\ln |\tan u + \sec u| \right]_0^{\frac{\pi}{3}}$$

$$= \left[\frac{2\sqrt{3}}{2} - 0 \right] - \frac{1}{2} \left[\ln |\sqrt{3} + 2| - \ln |0 + 1| \right]$$

$$= \sqrt{3} - \frac{1}{2} \ln(2 + \sqrt{3}) \quad \text{--- (1)}$$

a) $n=1$ LHS = $\sqrt{1} = 1$

RHS = $\frac{2}{3}\sqrt{1} = \frac{2}{3}$

$1 \geq \frac{2}{3} \checkmark$

$\Rightarrow$ true for $n=1$

Assume it holds true for all $n=k \geq 1$

so $\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{k} \geq \frac{2}{3}k\sqrt{k}$

to prove

$\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{k} + \sqrt{k+1} \geq \frac{2}{3}(k+1)\sqrt{k+1}$

LHS = $\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{k} + \sqrt{k+1}$

$\geq \frac{2}{3}k\sqrt{k} + \sqrt{k+1}$ — (1)

so to prove $\frac{2}{3}k\sqrt{k} + \sqrt{k+1} \geq \frac{2}{3}(k+1)\sqrt{k+1}$

Since everything is positive, same as proving
square of both & rearrange

to prove: $2k^{\frac{3}{2}} \geq (2k-1)\sqrt{k+1}$

$4k^3 \geq (2k-1)^2(k+1)$

RHS: $(2k-1)^2(k+1)$

$= 4k^3 - 3k + 1$

so need to prove

$-3k + 1 \leq 0$

i.e. $3k \geq 1$
 $k \geq \frac{1}{3}$

but $k \geq 1$ so all the above hold true

$$\Rightarrow \sqrt{1} + \sqrt{2} + \dots + \sqrt{k+1} \geq \frac{2}{3}(k+1)\sqrt{k+1}$$

b) 1). $|\vec{OA}| = |\vec{OB}| = 1$

• $\vec{OP} = 3\vec{PB}$

• $\vec{AQ} = 2\vec{QP}$

• O, Q, R collinear

* $\vec{AR} = k\vec{AB}$

* $\vec{OP} = \frac{3}{4}\vec{OB} = \frac{3}{4}\vec{b}$

* $\vec{OQ} = \vec{OA} + \vec{AQ}$

$$= \vec{OA} + \frac{2}{3}\vec{AP}$$

$$= \vec{OA} + \frac{2}{3}(\vec{OP} - \vec{OA})$$

$$= \vec{OA} + \frac{2}{3}\left(\frac{3}{4}\vec{OB} - \vec{OA}\right)$$

$$= \vec{a} + \frac{2}{3}\left(\frac{3}{4}\vec{b} - \vec{a}\right)$$

$$= \vec{a} + \frac{1}{2}\vec{b} - \frac{2}{3}\vec{a}$$

$$= \frac{1}{3}\vec{a} + \frac{1}{2}\vec{b} \quad \text{--- (1)}$$

* O, R, Q collinear so $\vec{OR} = \mu\vec{OQ}$
$$= \mu\left[\frac{1}{3}\vec{a} + \frac{1}{2}\vec{b}\right]$$

* Also $\vec{OR} = \vec{OA} + \vec{AR} = \vec{OA} + k\vec{AB}$

$$= \vec{OA} + k(\vec{OB} - \vec{OA})$$

$$= \vec{a} + k(\vec{b} - \vec{a})$$

$$= (1-k)\vec{a} + k\vec{b} \quad \text{--- (1)}$$

$$\text{but } \vec{OR} = \frac{1}{3}\mu\vec{a} + \frac{1}{2}\mu\vec{b}$$

$$\Rightarrow \frac{\mu}{3} = 1-k \quad \text{--- (1)}$$

$$\frac{\mu}{2} = k$$

$$\mu = 2k \quad \text{--- (2)}$$

sub (2) in (1)

$$\frac{2k}{3} = 1-k$$

$$\frac{5k}{3} = 1$$

$$k = \frac{3}{5} \quad \text{--- (1)}$$

$$\text{ii) } \vec{OQ} \cdot \vec{AP} = 0$$

$$\left(\frac{1}{3}\vec{a} + \frac{1}{2}\vec{b}\right) \cdot \left(\frac{3}{4}\vec{b} - \vec{a}\right) = 0$$

$$\frac{1}{4}\vec{a} \cdot \vec{b} - \frac{1}{3}|\vec{a}|^2 + \frac{3}{8}|\vec{b}|^2 - \frac{1}{2}\vec{a} \cdot \vec{b} = 0$$

$$-\frac{1}{4}\vec{a} \cdot \vec{b} + \frac{1}{24} = 0 \quad \left(\begin{matrix} |\vec{a}| = |\vec{b}| \\ = 1 \end{matrix}\right)$$

$$\frac{1}{4}\vec{a} \cdot \vec{b} = \frac{1}{24}$$

$$\vec{a} \cdot \vec{b} = \frac{1}{6} \quad \text{--- (1)}$$

$$\cos \angle AOB = \frac{a \cdot b}{|a||b|}$$

$$\cos \angle AOB = \frac{1}{6}$$

$$\angle AOB \approx 80^\circ 24' \quad \text{--- (1)}$$

$$c) a_{up} = -g - kv$$

$$\frac{dv}{dt} = -g - kv$$

$$\frac{dt}{dv} = \frac{1}{-g - kv}$$

$$\int_0^{t_{up}} dt = \int_u^0 \frac{dv}{-g - kv}$$

$$t_{up} = \frac{1}{k} \ln \left(\frac{g + ku}{g} \right)$$

$$a_{down} = g - kv$$

$$\frac{dv}{dt} = g - kv$$

$$\frac{dt}{dv} = \frac{1}{g - kv}$$

$$\int_0^{t_{down}} dt = \int_0^u \frac{dv}{g - kv}$$

$$t_{down} = \frac{1}{k} \ln \left(\frac{g}{g - ku} \right)$$

$$T_{total} = \frac{1}{k} \ln \left(\frac{g + ku}{g - ku} \right) \quad \text{--- (1)}$$

$$* -V \frac{dv}{dr} = -g - kv$$

$$\frac{dr}{dv} = \frac{-V}{g + kv}$$

$$r_{up} = \int_u^0 \frac{-V}{g + kv} dv$$

$$V \frac{dv}{dr} = g - kv$$

$$\frac{dr}{dv} = \frac{V}{g - kv}$$

$$r_{down} = \int_0^u \frac{V}{g - kv} dv$$

$$x_u = - \int_{\omega}^0 \frac{v}{g+kv} dv$$

$$x_D = - \int_0^{\omega} \frac{-v}{g-kv} dv$$

$$x_u = - \frac{1}{k} \int_{\omega}^0 \frac{kv + g - g}{g+kv} dv$$

$$x_D = - \frac{1}{k} \int_0^{\omega} \frac{-kv + g - g}{g-kv} dv$$

$$x_u = - \frac{1}{k} \int_{\omega}^0 \left[1 - \frac{g}{g+kv} \right] dv$$

$$x_D = - \frac{1}{k} \int_0^{\omega} \left[1 - \frac{g}{g-kv} \right] dv$$

$$x_u = \frac{\omega}{k} - \frac{g}{k^2} \ln \left(\frac{g+k\omega}{g} \right)$$

$$x_D = - \frac{\omega}{k} + \frac{g}{k^2} \ln \left(\frac{g}{g-k\omega} \right)$$

— (1)

$$x_{up} = x_D$$

$$\frac{\omega}{k} - \frac{g}{k^2} \ln \left(\frac{g+k\omega}{g} \right) = - \frac{\omega}{k} + \frac{g}{k^2} \ln \left(\frac{g}{g-k\omega} \right)$$

$$\frac{\omega + \omega}{k} = \frac{g}{k^2} \ln \left(\frac{g+k\omega}{g-k\omega} \right)$$

so $T = \frac{1}{k} \ln \left(\frac{g+k\omega}{g-k\omega} \right)$

$$\frac{g}{k} T = \frac{\omega + \omega}{k}$$

$$T = \frac{\omega + \omega}{g}$$

— (1)

d) Since $x \ln x$ is strictly increasing, the right-endpoint rectangles of width 1 will overestimate the area under curve

$$\sum_{k=1}^n k \ln k > \int_1^n x \ln x \, dx \quad \text{--- (1)}$$

$$\int_1^n x \ln x \, dx = \left[\frac{x^2}{2} \ln x \right]_1^n - \int_1^n \frac{x}{2} \, dx$$

$$= \frac{n^2}{2} \ln n - \left[\frac{x^2}{4} \right]_1^n$$

$$= \frac{n^2}{2} \ln n - \frac{n^2}{4} + \frac{1}{4}$$

$$= \frac{n^2}{4} (2 \ln 2 - 1) + \frac{1}{4} \quad \text{--- (1)}$$

so $\sum_{k=1}^n k \ln k > \int_1^n x \ln x \, dx$

$$(\ln(2^2 \times 3^3 \times \dots \times n^n)) > \frac{n^2}{4} (2 \ln n - 1) + \frac{1}{4}$$

$$\ln a_n > \frac{n^2}{4} (2 \ln n - 1) \quad \text{--- (1)}$$

e both sides $a_n > e^{\frac{n^2}{4} (2 \ln n - 1)} \quad \text{--- (1)}$